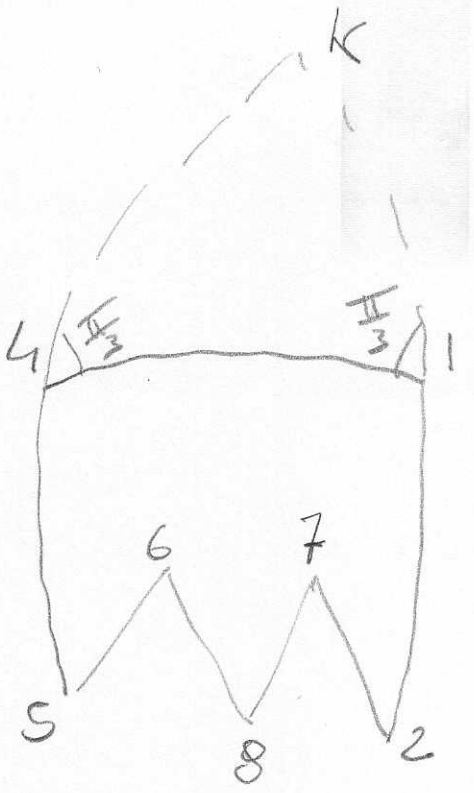
300

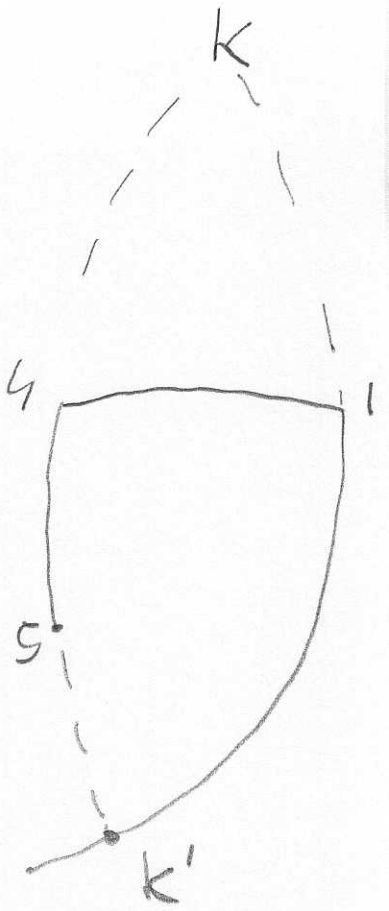
Observe that the net cannot contain an edge long π a nice because the sum of the normalized areas of its convex faces is 9

Consider the face 127'10'654.
If we remove the edges 6'10' and '10'7 we obtain a $\{7,2\}$ heptagon with two obtuse angles in 7 and 6 and three equal edges 54, 41, and 12 because these belong to the quadrilaterals '11'549, 9410, and 0123.
Moreover all the edges of the heptagon are shorter than π



Recall that 41 is at least long as the edge of the equilateral triangle with angles $\frac{2\pi}{3}$. Prolong the edges 45 and 51 until they meet in K. We then conclude that the angle in K is $\geq \frac{2\pi}{3}$

Let K' be the antipodal of K



At least one of the two edges 45 and 12 is shorter than $4k' = 1k'$

But $4k'$ and $1k'$ are both shorter than the edge of the equilateral triangle with angles $\frac{2\pi}{3}$ because the angle in k' is at least $\frac{2\pi}{3}$. So the edges 45, 12, and 14 cannot be all equal.